



Short communication

Exact solution of the cubic-quintic Duffing oscillator

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ABSTRACT

In this paper, we derive the exact solution of the cubic-quintic Duffing oscillator based on the use of Jacobi elliptic functions. We also showed that the exact angular frequency of this cubic-quintic Duffing equation is given in terms of the complete elliptic integral of the first kind.

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1. Introduction

Duffing oscillators are described by nonlinear differential equations that modeled the behavior of many practical problems that arise in engineering, physics, and in many real world applications [1,2]. It is well-known that Duffing oscillators that have strong third and fifth power nonlinear terms can be found in the modeling of free vibrations of a restrained uniform beam with intermediate lumped mass, the nonlinear dynamics of slender elastica, the generalized Pochhammer–Chree (PC) equation, the generalized compound KdV equation in nonlinear wave systems, among others [3]. Due to the presence of these strong nonlinearities, the derivation of accurate solutions of nonlinear Duffing oscillators becomes demanding. In spite of the difficulties involved, many researchers have used perturbation and non-perturbative techniques in an attempt to derive accurate approximate solutions of these strong nonlinear oscillators [4–10]. For instance, Wu et al. in [4] and Lai et al. in [9] developed analytical approximate solutions for a class of strongly non-linear oscillators based on the combination of Newton's method and the harmonic balance method which are valid for the entire range of oscillation amplitudes. By applying He's variational approach, Ganji et al. obtained the approximate solutions of the cubic-quintic Duffing oscillator with results that showed good agreement with its numerical integration solution [6].

Recently, some other techniques were applied to obtain approximate solutions to the cubic-quintic Duffing equation. These methods include the Homotopy Analysis Method (HAM) and the Homotopy Pade technique [7], He's Energy Balance Method (EBM) [8], a solution procedure of Coupled Method of Homotopy perturbation Method and Variational Method [11], and a modified iteration technique to solve strong nonlinear differential equations [3,12].

The aim of this paper is to derive the exact solution of a cubic-quintic Duffing oscillator of the form

$$\ddot{x} + Ax + Bx^3 + Gx^5 = 0, \quad (1)$$

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where, x denotes the displacement of the system, A , B and G are real system constant parameters. In order to obtain the exact solution of Eq. (1), we will make use of Jacobi elliptic functions. The main motivation in using elliptic functions comes from the fact that exact solutions of some nonlinear oscillators have been successfully found by using elliptic functions. For instance, Tamura [13], Rand [14], and Hu [15] used Jacobian elliptic functions to obtain the exact solution of some quadratic nonlinear oscillators. Also, Elías-Zúñiga used in [16] the incomplete elliptic integral of the second kind to obtain the exact solution of Lamé's equation. By using a periodic rational form Jacobi elliptic function, Elías-Zúñiga in [17] derived the exact solution of the quadratic mixed-parity Helmholtz–Duffing oscillator.

On the basis of these findings, we shall next derive the exact solution of the cubic–quintic Duffing oscillator (1) by using Jacobi elliptic functions.

2. Exact solution

In this section we shall derive the exact solution of Eq. (1) by assuming that this solution is given by the following equation:

$$x^2(t) = \frac{1}{a + b \operatorname{cn}^2(\omega t + \phi, k^2)}, \quad (2)$$

where $\operatorname{cn}(\omega t + \phi, k^2)$ is the cn Jacobian elliptic function that has a period in ωt equal to $4K(k^2)$, and $K(k^2)$ is the complete elliptic integral of the first kind for the modulus k . Here, a , b , k , ω , and ϕ are unknown parameters that need to be determined. We have assumed that the rational form elliptic function (2) represents the exact solution of Eq. (1), since we were able to determined the exact solution of nonlinear differential equations with quadratic and cubic terms. See [17] and references cited therein.

We next substitute Eq. (2) into Eq. (1) to get:

$$a^2A + aB + G - ab\omega^2 + abk^2\omega^2 + \operatorname{cn}^2(\omega t + \phi, k^2)[2aAb + bB + 2ab\omega^2 + 2b^2\omega^2 - 4abk^2\omega^2 - 2b^2k^2\omega^2] + \operatorname{cn}^4(\omega t + \phi, k^2)[Ab^2 - b^2\omega^2 + 3abk^2\omega^2 + 2b^2k^2\omega^2] = 0 \quad (3)$$

in which the following identities

$$\operatorname{sn}(\omega t + \phi, k^2)^2 + \operatorname{cn}(\omega t + \phi, k^2)^2 = 1; \quad \operatorname{dn}(\omega t + \phi, k^2)^2 + k^2\operatorname{sn}(\omega t + \phi, k^2)^2 = 1 \quad (4)$$

for the $\operatorname{sn}(\omega t + \phi, k^2)$ and $\operatorname{dn}(\omega t + \phi, k^2)$ Jacobian elliptic functions have been used.

Note that Eq. (3) holds for all time t if and only if:

$$a^2A + aB + G - ab\omega^2 + abk^2\omega^2 = 0, \quad (5)$$

$$2aAb + bB + 2ab\omega^2 + 2b^2\omega^2 - 4abk^2\omega^2 - 2b^2k^2\omega^2 = 0, \quad (6)$$

$$Ab^2 - b^2\omega^2 + 3abk^2\omega^2 + 2b^2k^2\omega^2 = 0. \quad (7)$$

Thus, we have three algebraic equations and five unknowns parameters a , b , k , ω , and ϕ . To determine the value of these parameters, we also need to use the initial conditions of Eq. (1) that without loss of generality, are assumed to be given as:

$$x(0) = x_{10}, \quad \dot{x}(0) = 0. \quad (8)$$

Thus, by using Eq. (2) and the initial conditions given by Eq. (8), we get that $\phi = 0$ and

$$b = \frac{1 - ax_{10}^2}{x_{10}^2}. \quad (9)$$

By solving Eqs. (6) and (7), we obtain after some algebraic manipulations that:

$$k^2 = \frac{(2A + Bx_{10}^2 - 2a^2Ax_{10}^4 - aBx_{10}^4)}{2A + 4aAx_{10}^2 + 2B^2x_{10}^2 + aBx_{10}^4}, \quad (10)$$

$$\omega^2 = -\frac{(2A + 4aAx_{10}^2 + 2Bx_{10}^2 + aBx_{10}^4)}{2(1 + ax_{10}^2 + a^2x_{10}^4)}. \quad (11)$$

We next substitute the expressions of k and ω given by Eqs. (10) and (11) into Eq. (5), to get:

$$a^2(6A + 3Bx_{10}^2 + 2Gx_{10}^4) + a(3B + 2Gx_{10}^2) + 2G = 0, \quad (12)$$

that has the solutions

$$a = -\frac{4G}{3B + 2Gx_{10}^2 \pm \sqrt{3}\sqrt{-16AG + (B - 2Gx_{10}^2)(3B + 2Gx_{10}^2)}}. \quad (13)$$

Therefore, the exact solution of Eq. (1) is given by

$$x^2(t) = \frac{1}{a + b \operatorname{cn}^2(\omega t, k^2)} \quad (14)$$

in which the values of a , b , k , and ω can be determined from Eqs. (9)–(11) and Eq. (13) for given parameters values of A , B , and G .

Since the Jacobian elliptic function $\operatorname{cn}(\omega t, k^2)$ has a period in ωt equal to $4K(k^2)$, we may see that the corresponding exact period of oscillation T of the cubic–quintic Duffing nonlinear oscillator described by Eq. (1) is given by

$$T = \frac{4K(k^2)}{\omega}. \quad (15)$$

Notice that during the derivation of the exact solution of the cubic–quintic Duffing oscillator (1), we have not imposed any restrictions to the argument and modulus values of the Jacobi elliptic functions. Therefore, depending on the system parameter values of A , B , and G , we can have real or imaginary values for a , b , k , and ω [18]. For instance, let us assume the following system parameter values of $A = 1$, $B = 1$, $G = 1$, and $x(0) = 1$. Then, by using Eqs. (9)–(11), we obtained that $a = -0.227273 \pm 0.360784i$, $b = 1.22727 \mp 0.360784i$, $k = 0.959403 \pm 0.282038i$, $\omega = 0.218183 \pm 1.51578i$, where $i = \sqrt{-1}$. We next compare the amplitude–time response curve obtained by numerical integration of Eq. (1) with our derived exact solution described by Eq. (14). It is clearly seen from Fig. 1 that both solutions are the same and have the exact circular frequency value of 1.52359 [8]. In this Fig. 1, the solid gray line represents our derived exact solution while the red dashed line represents the numerical integration solution. If we next use the parameter values of $A = 2$, $B = 1$, $G = 1$, and $x(0) = 5$, we get from Eq. (15) the circular frequency value of 19.2104 which match exactly with the numerical integration value [8]. Fig. 2 shows the angular frequency values obtained by using Eq. (15) plotted versus the initial displacement x_{10} . Here, we have used the following system parameter values: $A = 1$, $B = 5$, $G = 10$. We may see from Fig. 2, that the angular frequency values obtained from Eq. (15) follow exactly the numerical integration solutions of Eq. (1).

To further assess the accuracy of our derived solution (13), we shall next compare its theoretical predictions with the solution to Eq. (1) derived by Maimistov in [2]:

$$x^2(\tau) = \frac{2a_1}{\mu + \sqrt{1 + 4a_1\varepsilon} \cosh[2a_1^{1/2}(\tau)]}, \quad (16)$$

where

$$a_1 = -A; \quad \mu = \frac{B}{2}; \quad \varepsilon = \frac{G}{3}. \quad (17)$$

Fig. 3 shows a comparison of the soliton profiles obtained by Eq. (16) with the numerical integration solution obtained by using the Fourth order Runge–Kutta method. As we may see from Fig. 3(a), for long periods of time, solution (16) does not follow the numerical integration solution. This is mainly due to the fact that the hyperbolic function $\cosh[2a_1^{1/2}(\tau)]$ in Eq. (16), is not a periodic function when $a_1 > 0$. Fig. 3(b) shows theoretical predictions obtained by using our derived solution (14) and the numerical integration response curves of Eq. (1). In this case, both solutions agree for the entire time span showed in Fig. 3(b).

Also, we have found that the soliton-like solution behavior of Eq. (1) arises when the factor of the quadratic term in (12) becomes zero i.e., when

$$6A + 3Bx_{10}^2 + 2Gx_{10}^4 \equiv 0. \quad (18)$$

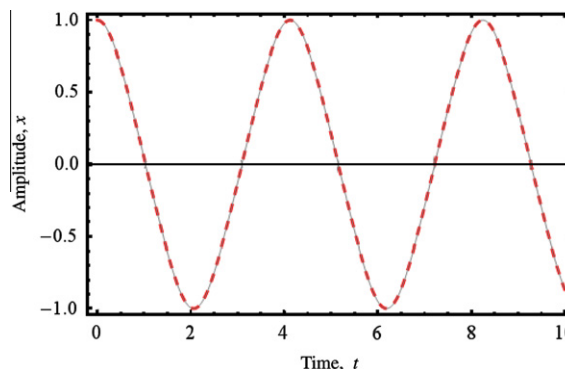


Fig. 1. Comparison of our derived exact solution with the numerical integration solution of the cubic–quintic Duffing oscillator equation for the system parameter values of $A = 1$, $B = 1$, $G = 1$, $x(0) = 1$, and $\dot{x}(0) = 0$. Here, the solid gray line represents our derived exact solution while the the red dashed lines represent the numerical integration solution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

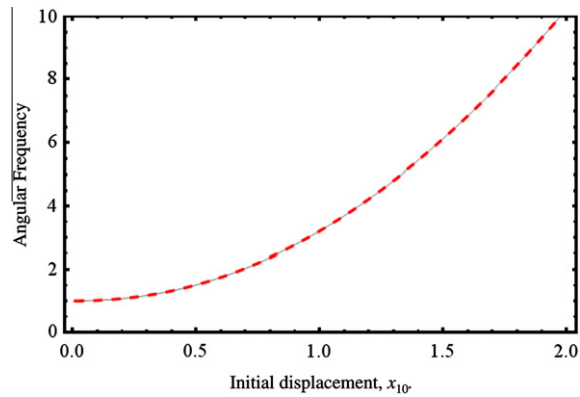


Fig. 2. Angular frequency values plotted versus the initial condition x_{10} . The system parameter values are $A = 1$, $B = 5$, and $G = 10$. Here, the solid gray lines represent theoretical predictions of the angular frequency values obtained from Eq. (15) while the red dashed lines represent the numerical integration angular frequency values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

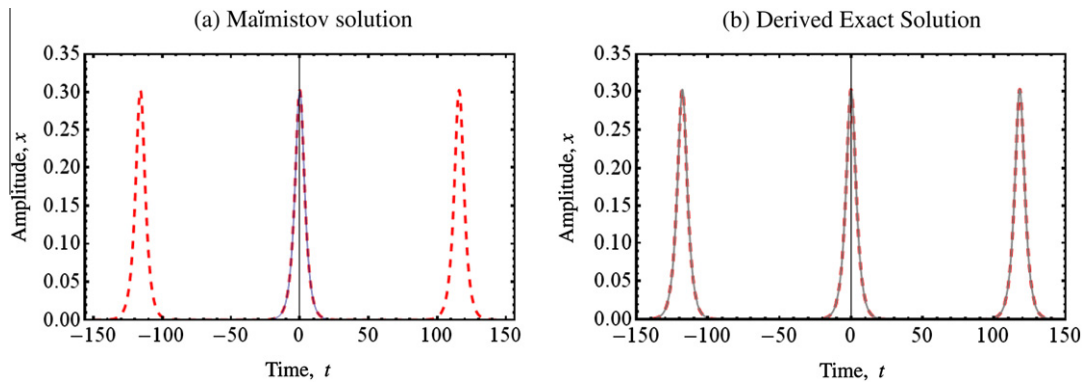


Fig. 3. Comparison of soliton profiles obtained by numerical integration of Eq. (1) with (a) Maimistov solution, and (b) our derived exact solution. The system parameter values are $a_1 = 0.1$, $\mu = 1$, $\varepsilon = 1$, $x(0) = 0.30266$, and $\dot{x}(0) = 0$. By using Equations (9)–(11), we have obtained that $a = -0.91608$, $b = 11.8322$, $k = 0$, $\omega = 0.3162i$, where $i = \sqrt{-1}$. Here, the solid gray lines represent our derived exact soliton solution while the red dashed lines represent the numerical integration solutions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

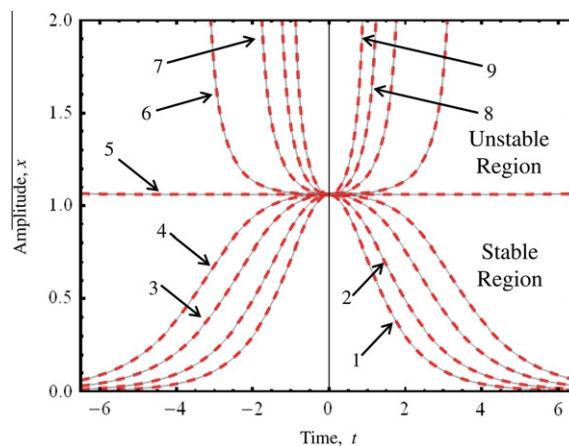


Fig. 4. Comparison of soliton profiles obtained by numerical integration of Eq. (1) with our derived exact solution given by Eq. (14). In all cases, the value of $k = 0$ and the computed values for A, G, a, b , and ω are shown in Table 1, where $i = \sqrt{-1}$. Here, the solid gray lines represent our derived exact solutions while the red dashed lines represent the numerical integration solutions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Computed parameter values of the curves showed in Fig. 3.

Curve	G	A	a	b	ω
1	−0.3	−1	0.1127	0.7745	$0 + 1i$
2	−0.9	−0.7459	0.4532	0.4340	$0 + 0.8636i$
3	−1.2	−0.6189	0.7283	0.1589	$0 + 0.7867i$
4	−1.29	−0.5808	0.8343	0.0529	$0 + 0.7621i$
5	−1.33	−0.5636	0.8867	0.0005	$0 + 7507i$
6	−1.35	−0.5554	0.9130	−0.0257	$0 + 0.7452i$
7	−1.5	−0.4919	1.1455	−0.2581	$0 + 0.7013i$
8	−1.8	−0.3649	1.8530	−0.9657	$0 + 6040i$
9	−2.5	−0.1108	8.1311	−7.2438	$0 + 3329i$

Results from theoretical predictions by considering the relation (18) are now showed in Fig. 4. Here, we have assumed the following system parameter values of $B = 2$, $x(0) = x_{10} = 1.06161$, $\dot{x}(0) = 0$, and we have used relation (18) to obtain the corresponding values of A for assigned values of G , as showed in Table 1. From Fig. 4, we may see that theoretical predictions obtained from the exact solution given by Eq. (14) coincide with those of the numerical integration solutions. Also, notice from Fig. 4 that for parameter values of $A \geq -0.5635$, the system response becomes unstable.

As expected, our exact solution of Eq. (1) works for any amplitude of oscillation x_{10} and for any given parameter values of A , B , and G . See for instance [9,19] and references cited there in.

3. Some concluding remarks

In this paper we have presented the exact solution of the cubic-quintic Duffing Eq. (1) that involve the usage of Jacobi elliptic functions. One must bear in mind that depending on the system parameters values of A , B , G , we can have Jacobi elliptic functions with real or complex argument and modulus values and thus, we may need to apply Jacobi elliptic functions imaginary identities to obtain the amplitude-time response curve of $x(t)$ given by Eq. (14) [18]. Also, we may notice that when the modulus $k = 1$, Eq. (14) can be written in equivalent form as:

$$x^2(t) = \frac{1}{a + b \operatorname{sech}^2(\omega t)}, \quad (19)$$

which corresponds to the exact soliton solution for derivative nonlinear Schrödinger equation that satisfies certain value conditions of the system parameters A , B , and D [20–22]. Of course, our derived exact solution can satisfy any parameter value condition and thus, we may obtain a great variety of exact soliton solutions of the derivative nonlinear Schrödinger equation that model the physical behavior of many real world problems of practical applications. Also, we have found that the soliton-like behavior of Eq. (1) occurs if and only if Eq. (18) is satisfied.

Finally, one must bear in mind that the exact solution of Eq. (1) can be used in conjunction with perturbative or nonperturbative techniques to derive approximate solutions of nonlinear oscillators whose closed-form solutions are unknown [23–26]. Future work is focused on assuming solutions of the form (14) to investigate exact or approximate solutions of cubic-quintic, damped nonlinear systems that modelled the behavior of many engineering systems with finite degrees of freedom.

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